

Open Source MLOps Tools: Ideal for Managing ML Data Workflows

MLOps adds automation, organisation and reliability to the machine learning lifecycle. Open source MLOps tools do a great job of helping build a machine learning model, with each tool tackling a distinct challenge.



When building a machine learning (ML) model what really makes the difference is the quality with which the underpinning data pipelines are architected, operated and aligned to the needs of the model. If the behind-the-scenes processes such as data preparation, monitoring or workflow coordination are variable or poorly executed, even the best algorithms and most capable teams may fail to deliver. The continuous flow of activities from data gathering and preprocessing through version control, model

training and evaluation, deployment, and post-deployment monitoring is key to a reliable machine learning system. Failure of a project, wastage of resources, or delays can be caused by any single point of failure in this chain. This is addressed by MLOps, which introduces automation and definition into the process, making it more transparent and uniform. While most teams cannot meet the high costs of commercial MLOps platforms, open source technology is redefining the space. These flexible, budget-friendly solutions are making it possible for

companies of all sizes to manage and scale machine learning projects, driving innovation and pushing AI research for industries everywhere.

Why MLOps is the backbone of sustainable machine learning

Machine learning operations, or MLOps, adds structure to the frequently disorganised process of creating and implementing machine learning models. It is more than just a collection of tools. While traditional machine learning often emphasises getting the model's prediction right, MLOps